TEAM-CONNECT

SaaS HR & Workforce Management Platform

System Architecture, Data Model & AI Integration

Version 2 — Work Reports, Underperformance Warnings & AI Insights

1. Purpose & Product Positioning

Team-Connect is a multi-tenant, offline-first HR and workforce management SaaS purpose-built for field and deskless teams operating in low-connectivity environments. Beyond standard HR functions, it introduces a daily Work Report confirmation workflow that converts field activity into verifiable, GPS-checked pay days, automatic underperformance detection and warning letters, and an AI Insights layer that scores performance, forecasts hiring needs, and surfaces skill gaps.

The platform is structured as fifteen modules across four tiers — Core & Administrative, Talent & Growth, Advanced & Support, and Intelligence — sold as a tiered subscription (Starter / Growth / Enterprise / Intelligence) on a single shared multi-tenant core.

2. SaaS Foundation Principles

- Tenant isolation: every table carries a tenant_id, enforced with row-level security in PostgreSQL rather than app-level filtering alone.
- Canonical employee record: one employees table that every module references by employee_id.
- Entitlements engine: a tenant_modules table gates API access per tenant per module.
- Event-driven decoupling: modules communicate through an internal event bus (Redis pub/sub + job queue) rather than querying each other's tables directly.
- Offline-first field layer: the mobile app and SMS/USSD channel queue actions locally and sync opportunistically, with server-side conflict resolution.
- Confirmation-gated pay: a field day only counts toward salary once explicitly confirmed by an admin — self-reported data alone never triggers payment.

3. System Architecture

The diagram adds two new always-on services to the original design: an AI Insights Engine that consumes events from Attendance, Work Reports, Performance, and LMS to generate scores and recommendations, and a Notifications & Letters Engine that watches for underperformance and automatically generates and delivers warning letters.

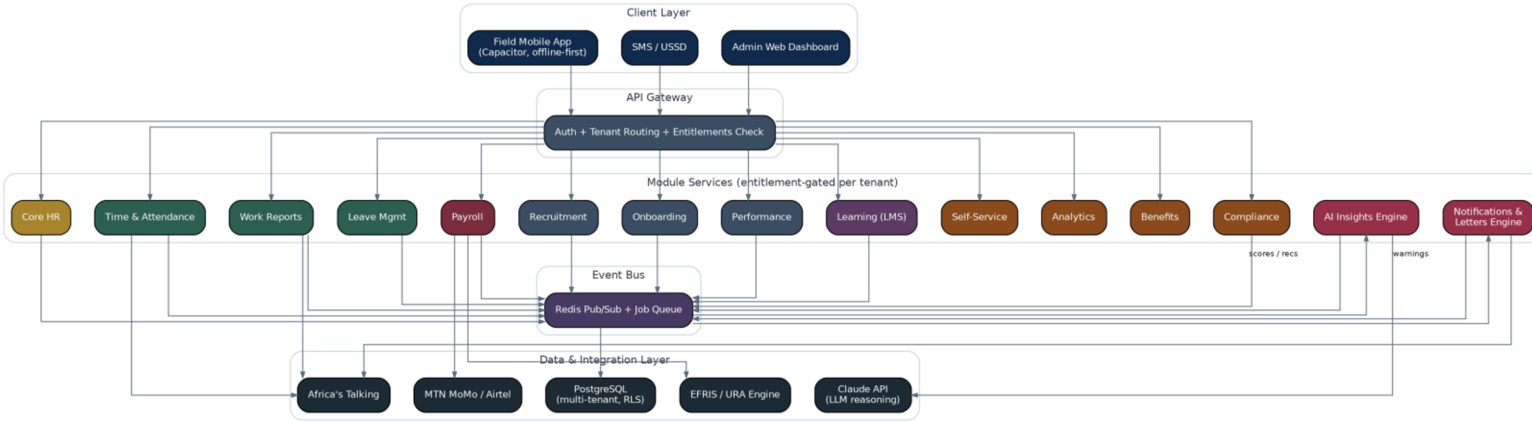

Figure 1 — Team-Connect system architecture, including Work Reports, AI Insights Engine, and Notifications & Letters Engine.

3.1 Layer Notes

AI Insights Engine

Subscribes to events from Work Reports, Attendance, Performance, and LMS. Calls the Claude API to reason over aggregated employee and department data, then writes structured results back to dedicated ai_* tables (see Section 5.14). It never writes directly to payroll or core employee records — its outputs are recommendations and scores for a human admin to act on.

Notifications & Letters Engine

Watches monthly_hour_targets and performance_warnings for threshold breaches, generates warning or underperformance letters from templates, and delivers them via SMS/email through Africa's Talking. It logs every letter it produces in generated_letters for audit purposes.

4. Module Catalogue & Subscription Tiers

Modules are grouped into four functional tiers. Work Reports ships in Starter since it is core to field-team pay integrity. AI Insights and the Notifications & Letters engine are grouped into a new Intelligence tier, sold as an add-on to Growth or bundled into Enterprise.

Tier	Modules Included	Target Customer
Starter	Core HR, Time & Attendance, Work Reports, Leave Management, Employee Self-Service	Small field crews and SMEs — matches Softnodes' own current scale
Growth	+ Payroll, Recruitment (ATS), Onboarding	SMEs scaling headcount and formalising hiring
Enterprise	+ Performance Management, LMS, Analytics, Benefits Administration, Compliance	Larger organisations and government contractors requiring audit trails
Intelligence (add-on)	+ AI Insights Engine, Notifications & Letters Engine	Tenants wanting automated performance scoring, hiring forecasts, skill-gap analysis, and auto-generated warning letters

Table 1 — Subscription tiers and module inclusion.

4.1. Tenant (Company) Sign-Up — self-serve

1. Company admin registers: company name, TIN, phone/email, picks a tier (Starter/Growth/Enterprise)
2. OTP verification via SMS (Africa's Talking) or email
3. Creates the `tenants` row + one SuperAdmin `accounts` row
4. Optional trial period (e.g. 14 days) before MoMo/card billing kicks in
5. Setup wizard: add departments → add first site(s) with geofence → invite first employees

This is the only place a password-style login makes sense — office admins have laptops and normal connectivity.

4.2. Employee Account Creation — two paths

a) Admin-initiated invite (default path) Admin adds the worker's record in Core HR with just a phone number → system SMS's a one-time signup code → worker opens the app (or dials USSD) → sets a 4–6 digit PIN → account is active. This matches the "under 2 minutes, phone number only" principle from the architecture doc — no email required.

b) Self-registration via site code (bulk onboarding) For situations like enrolling 20 day-laborers at once on a new site, the admin generates a **site enrollment code**. Workers enter the code + their phone number themselves → account sits in "pending" → admin approves in bulk → activated. Much faster than pre-entering every record one by one.

4.3. Authentication model

User type	Credential	Why
Field worker (app)	Phone number + PIN	Fast, low-literacy friendly, works with cheap Android
Field worker (USSD)	Phone number + PIN entered at each session	No persistent session — transaction-based, works with zero data
Office admin / HR / Payroll	Email + password (SSO later)	Higher-sensitivity access to pay and personal data warrants stronger auth

PINs are hashed (never stored plain), attempts are rate-limited, and a device-bound offline token lets the app validate the PIN locally for up to 48 hours without a network round-trip — consistent with the offline-first requirement.

4.4. Roles & permissions

- **SuperAdmin** — full tenant control, billing
- **HR Admin** — Core HR, Leave, Onboarding, Compliance

- **Site/Field Manager** — scoped to their assigned site(s) only: confirms work reports, approves leave for their crew
- **Payroll Admin** — separated out deliberately; not every HR admin should see disbursement data
- **Recruiter** — ATS only
- **Employee/Field Worker** — self-service only: their own work reports, payslips, leave requests

Permissions are checked at the API gateway against both the role **and** the tenant's active `tenant_modules` — so a Field Manager on a Starter-tier tenant simply won't see Performance Management options because that module isn't purchased, not because of a role hack.

4.5. Account lifecycle

```
Invited → Active → Suspended → Deactivated
```

On termination, the account is deactivated (locked out immediately, sessions revoked) but the `employee_id` and historical records stay intact — payroll history, work reports, and compliance flags must survive for audit even after someone leaves.

4.6 Data model additions

Table	Key Fields	Purpose
`accounts`	id, tenant_id, employee_id (nullable), phone, email, pin_hash/password_hash, role, status, last_login_at	Login identity, separate from the HR `employees` record
`invites`	id, tenant_id, employee_id, phone, code, expires_at, status	One-time signup codes sent by SMS
`site_enrollment_codes`	id, tenant_id, site_id, code, expires_at, max_uses, uses_count	Bulk self-registration codes per site
`otp_verifications`	id, account_id, code, purpose, expires_at, verified_at	OTP for signup, PIN reset, sensitive actions
`auth_sessions`	id, account_id, device_id, token, created_at, expires_at, offline_expiry	Session + offline-token tracking, revocable on termination

5. Module Detail & Database Tables

Each module below lists the database tables it owns. All tables carry tenant_id directly, or inherit tenant scope through a parent foreign key (e.g. work_report_slots inherits scope via work_report_id).

5.1 Core HR / Employee Database

The canonical worker record referenced by every other module.

Table	Key Fields	Purpose
employees	id, tenant_id, name, national_id, employment_type, day_rate, monthly_salary, nssf_no, tin, status	One row per worker; source of truth for identity and pay basis
departments	id, tenant_id, name	Organisational grouping used for reporting and AI hiring recommendations
employee_documents	id, employee_id, doc_type, file_url, expiry_date	Certifications, licenses, ID copies — expiry feeds Compliance
employment_contracts	id, employee_id, contract_type, start_date, end_date, pay_basis, rate	Contract history, supports salaried, day-rate, and piece-rate workers

5.2 Time & Attendance

GPS/geofence clock-in and shift scheduling, feeding both Payroll and Work Reports.

Table	Key Fields	Purpose
sites	id, tenant_id, name, geofence_lat, geofence_lng, geofence_radius_m	Job site definitions used for GPS matching across Attendance and Work Reports
shift_schedules	id, tenant_id, employee_id, site_id, date, planned_start, planned_end	Planned assignment of a worker to a site on a given date
attendance_events	id, employee_id, site_id, type, timestamp, gps_lat, gps_lng, gps_match, channel, sync_status	Individual clock-in/out events; channel can be app, SMS, or USSD

5.3 Work Reports (Daily Task Confirmation)

The core field-pay-integrity module. Each worker fills a daily report broken into five time slots. A submitted report immediately earns a provisional day_value of 0.5; the worker can edit it for 24 hours, after which it locks automatically. An admin then reviews the locked report against GPS and task detail and confirms it — raising day_value to 1.0 and counting it toward the month — or rejects it, which counts as 0 and can trigger a compliance flag.

Time Slots

Slot	Window	Net Hours	Counts Toward
1	08:00 – 10:00 (Morning)	2.0 h	Core daily target (8h)
2	10:00 – 12:00 (Late Morning)	2.0 h	Core daily target (8h)
3	12:00 – 15:00 (Afternoon)	2.0 h net (1h reserved for lunch/break)	Core daily target (8h)
4	15:00 – 18:00 (Late Afternoon)	2.0 h	Core daily target (8h)
5	18:00 – 22:00 (Evening)	up to 4.0 h	Overtime — tracked separately, configurable per tenant

Table 2 — Slot 1–4 sum to an 8-hour core workday, matching the 8 × 30 monthly baseline. Slot 5 is overtime, counted separately.

Confirmation & Pay State Machine

Submission is time-stamped; a background job compares now() to edit_deadline (submitted_at + 24h) and flips status to locked automatically — the worker cannot edit past that point regardless of app state. Confirmation is a deliberate admin action, never automatic.

Figure 2 — Work report lifecycle: draft → submitted (0.5) → locked → confirmed (1.0) or rejected (0).

GPS Tally Logic

- Within geofence radius of the assigned site → slot flagged “match”, admin confirms with confidence.
- Outside geofence radius → slot flagged “mismatch”, shown to admin before confirmation, not auto-rejected (rural GPS drift is expected).
- No GPS captured (device offline) → flagged “unavailable”, falls back to admin judgment, syncs later via the offline queue.

Monthly Hours, Underperformance Warnings & Auto-Letters

The system computes a running baseline of 8 hours per working day (8 × 30 by default, configurable per tenant). Confirmed and submitted hours are tracked separately so shortfalls are caught early, before month-end, while pay is always calculated from confirmed hours only.

- Daily job compares accumulated hours to the prorated target (8h × working days elapsed).
- If accumulated hours fall behind a configurable buffer (e.g. 90% of the prorated target), a warning notification is sent to the employee and their manager immediately.
- If the shortfall persists past a harder threshold (e.g. confirmed hours < 80% of the monthly target at period close), the Notifications & Letters Engine auto-generates an underperformance letter from a template, logs it, and delivers it by SMS/email — no manual drafting required.
- Repeated shortfalls can additionally raise a compliance_flags entry for HR follow-up, independent of the pay calculation.

Table	Key Fields	Purpose
work_reports	id, employee_id, site_id, report_date, status, day_value, total_hours, submitted_at, edit_deadline, locked_at, confirmed_by, confirmed_at	One row per worker per day; status drives pay eligibility
work_report_slots	id, work_report_id, slot_no, time_window, task_description, gps_lat, gps_lng, gps_match, hours	One row per filled time slot within a daily report
monthly_hour_targets	id, employee_id, period, target_hours, submitted_hours, confirmed_hours, status	Running tally per employee per pay period, drives warning triggers
performance_warnings	id, employee_id, period, reason, hours_short, letter_id, created_at	Log of every warning raised, linked to the letter it generated
generated_letters	id, tenant_id, employee_id, type, content, generated_at, delivered_channel	Audit trail of every auto-generated letter (underperformance, termination notice, commendation)

5.4 Leave Management

Accrual, balances, and approval workflow, cross-checked against site coverage in Time & Attendance.

Table	Key Fields	Purpose
leave_types	id, tenant_id, name, annual_entitlement_days	Configurable leave categories (annual, sick, compassionate, etc.)
leave_balances	id, employee_id, leave_type_id, year, days_accrued, days_used	Per-employee, per-year balance tracking
leave_requests	id, employee_id, leave_type_id, start_date, end_date, status, approved_by	Individual leave applications and their approval state

5.5 Payroll

Consumes confirmed days/hours from Work Reports and Attendance; disburses via mobile money.

Table	Key Fields	Purpose
payroll_runs	id, tenant_id, period, status	One row per payroll cycle per tenant
payroll_inputs	id, payroll_run_id, employee_id, confirmed_days, confirmed_hours, source	Snapshot of confirmed work-report days and hours feeding a specific run
payslips	id, payroll_run_id, employee_id, gross, paye, nssf, net, momo_txn_id	Per-employee output of a payroll run, including disbursement reference
tax_rules	id, tenant_id, rule_type, parameters_json, effective_date	PAYE bands, NSSF rate, LST — versioned so history stays auditable

5.6 Recruitment (ATS)

Job posting and applicant pipeline tracking.

Table	Key Fields	Purpose
job_postings	id, tenant_id, title, department_id, status	Open roles advertised by the tenant
applicants	id, job_posting_id, name, phone, resume_url, stage	Candidates against a posting and their current pipeline stage
applicant_stage_history	id, applicant_id, stage, changed_at, changed_by	Audit trail of pipeline movement

5.7 Onboarding

Checklist-driven guidance for new hires, including safety-certification gating before first clock-in.

Table	Key Fields	Purpose
onboarding_checklists	id, tenant_id, name, applies_to_role	Checklist templates, optionally role-specific
onboarding_tasks	id, checklist_id, title, required	Individual tasks within a checklist
employee_onboarding_progress	id, employee_id, task_id, status, completed_at	Per-employee completion state of each task

5.8 Performance Management

Goal-setting, reviews, and feedback, cross-linked to Work Reports for field roles.

Table	Key Fields	Purpose
performance_goals	id, employee_id, title, target_date, status	Individual goals and their progress state
performance_reviews	id, employee_id, reviewer_id, period, score, comments	Formal review cycle outcomes
feedback_entries	id, employee_id, submitted_by, type, content, submitted_at	Peer, manager, or self feedback entries

5.9 Learning & Development (LMS)

Course catalogue and skill tracking, feeding both Scheduling and the AI skill-gap analysis.

Table	Key Fields	Purpose
courses	id, tenant_id, title, skill_tag	Course catalogue entries
course_completions	employee_id, course_id, completed_at, score	Completion records per employee
skills	id, tenant_id, name, category	Skill taxonomy used across LMS, Scheduling, and AI Insights

Table	Key Fields	Purpose
employee_skills	employee_id, skill_id, proficiency_level, verified_by	Verified skill levels per employee

5.10 Employee Self-Service (ESS)

Mobile-first access to payslips, leave, and personal details.

Table	Key Fields	Purpose
notifications	id, tenant_id, employee_id, type, message, read_status, created_at	In-app/SMS notifications, including hour-shortfall warnings

5.11 Analytics & Reporting

Standard HR metrics plus contract-linked labour analytics.

Table	Key Fields	Purpose
report_snapshots	id, tenant_id, report_type, period, data_json, generated_at	Cached, pre-computed report data for fast dashboard loads

5.12 Benefits Administration

NSSF, medical insurance linkage, and allowances.

Table	Key Fields	Purpose
benefit_plans	id, tenant_id, name, type, provider	Available benefit plans per tenant
employee_benefits	id, employee_id, benefit_plan_id, enrolled_date, status	Per-employee enrolment records

5.13 Compliance

Statutory and certification compliance tracking, including patterns raised from Work Reports.

Table	Key Fields	Purpose
compliance_rules	id, tenant_id, rule_name, category, threshold_json	Configurable compliance rules (leave minimums, cert expiry windows, GPS-mismatch patterns)
compliance_flags	id, employee_id, rule_id, type, due_date, status	Individual flags raised against an employee, can block scheduling

5.14 AI Insights Engine

A tenant-scoped analysis layer, not a separate source of truth. It reads from Work Reports, Attendance, Performance, LMS, and Compliance via the event bus, calls the Claude API to reason over the aggregated data, and writes structured, explainable outputs back into its own tables for an admin to review and act on.

What it produces

- Performance scores — per employee, per period, derived from confirmed work-report rate, GPS-match consistency, attendance punctuality, and review history.
- Hiring recommendations — per department, comparing site coverage/workload trends against current headcount to flag under- or over-staffing before it becomes a delivery risk.
- Skill-gap analysis — per department, comparing skills required by active/upcoming contracts (e.g. an increase in CCTV installs) against verified employee_skills, to recommend specific LMS courses or hires.
- Every output is traceable to the model run that produced it, via ai_model_runs, so recommendations can be audited or rolled back.

Table	Key Fields	Purpose
ai_performance_scores	id, tenant_id, employee_id, period, score, inputs_summary_json, generated_at	Explainable performance score per employee per period
ai_hiring_recommendations	id, tenant_id, department_id, recommended_headcount, reasoning_summary, generated_at	Headcount recommendation with a plain-language rationale
ai_skill_gap_analysis	id, tenant_id, department_id, skill_id, gap_severity, recommended_action, generated_at	Per-department, per-skill gap severity and suggested response
ai_model_runs	id, tenant_id, model_name, run_type, input_ref, output_ref, run_at, status	Audit log of every AI analysis run, for traceability

5.15 Notifications & Letters Engine

A shared service used by Work Reports (underperformance letters) and, longer term, other modules needing templated, auto-delivered communication (onboarding reminders, compliance expiry notices). It reuses generated_letters and notifications from the tables already defined above rather than introducing a parallel schema.

6. Entity-Relationship Model

Split into two diagrams for readability. Diagram A covers the core workforce and pay path: Core HR, Attendance, Work Reports, Leave, and Payroll — the modules a Starter-tier tenant uses daily.

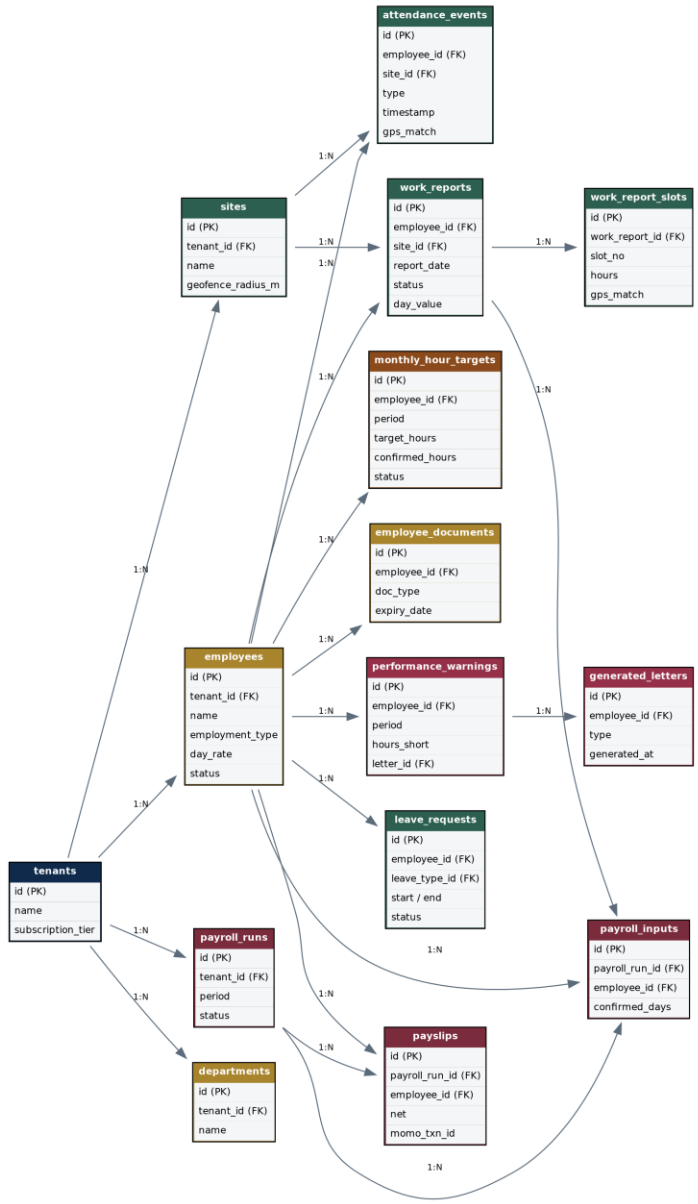

Figure 3 — Diagram A: Core HR, Attendance, Work Reports, Leave & Payroll.

Diagram B covers Talent & Growth, Advanced & Support, and the new AI Insights tables — the modules added at Growth, Enterprise, and Intelligence tiers.

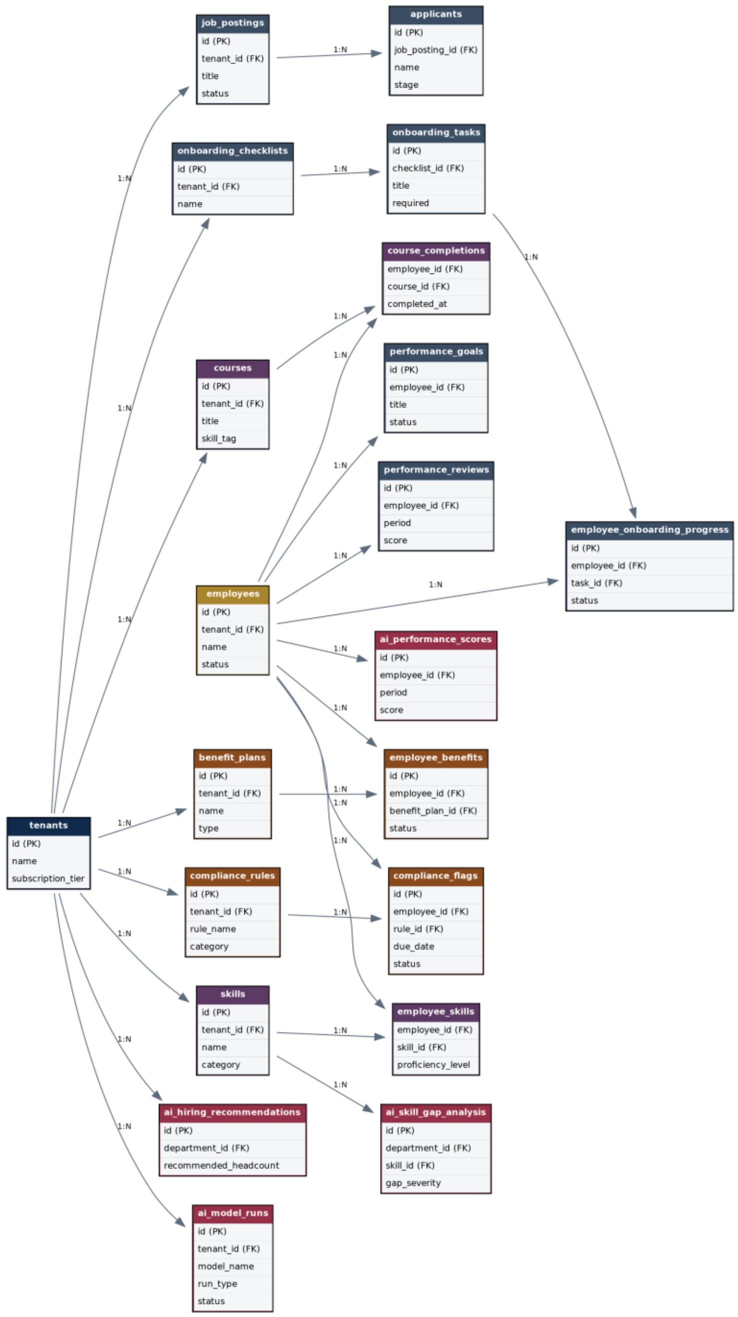

Figure 4 — Diagram B: Recruitment, Onboarding, Performance, LMS, Benefits, Compliance & AI Insights.

7. Recommended Build Sequence

Sequenced so each phase can be piloted on Softnodes' own field crews before being sold externally:

- Phase 1 — Core HR + Time & Attendance (offline sync, GPS geofence, SMS/USSD fallback).
- Phase 2 — Work Reports (slot filling, 24h lock, admin confirmation, GPS tally) — the highest-priority differentiator.
- Phase 3 — Monthly hour targets + underperformance warnings + auto-generated letters.
- Phase 4 — Leave Management + Employee Self-Service.
- Phase 5 — Payroll + MTN MoMo / Airtel Money disbursement, wired to confirmed Work Report days.
- Phase 6 — Onboarding + Recruitment (ATS).
- Phase 7 — Performance, LMS, Analytics, Compliance, Benefits (Enterprise tier).
- Phase 8 — AI Insights Engine (performance scoring, hiring recommendations, skill-gap analysis) + Notifications & Letters Engine generalisation (Intelligence tier).